



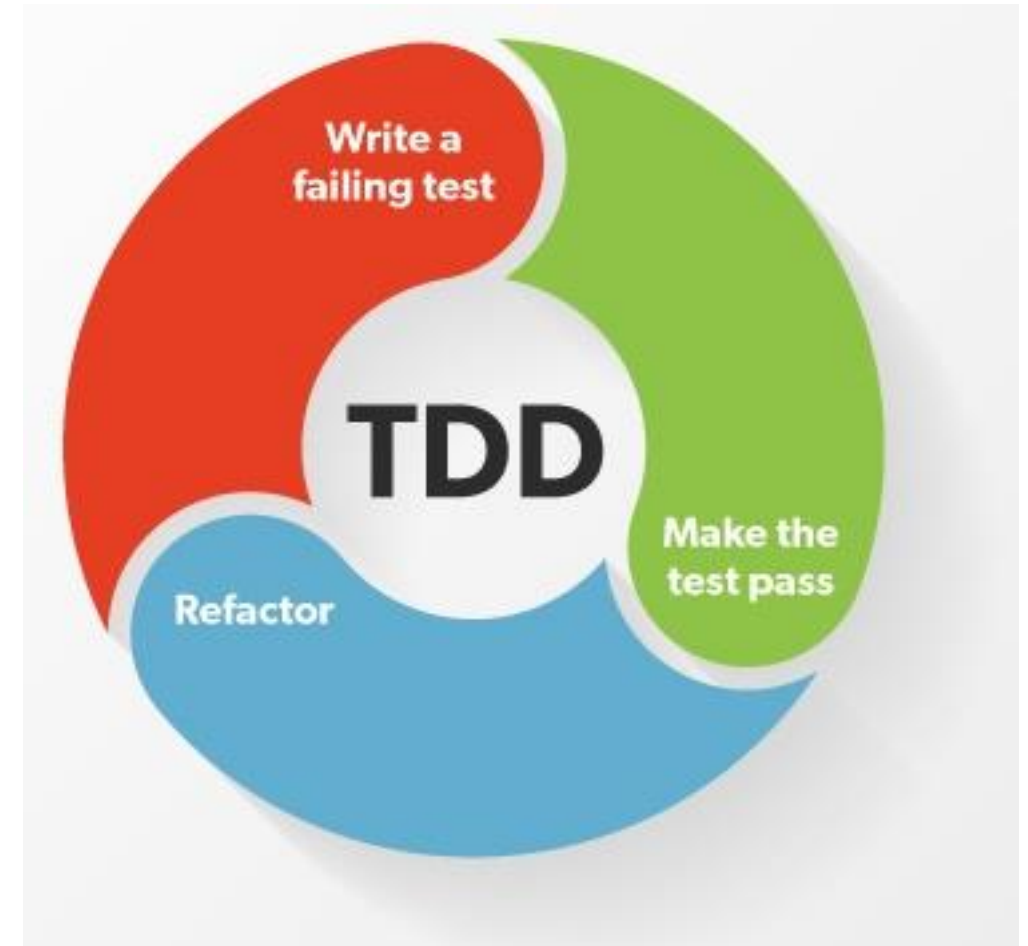
COM410 Programming in Practice

T5. Test Driven Development



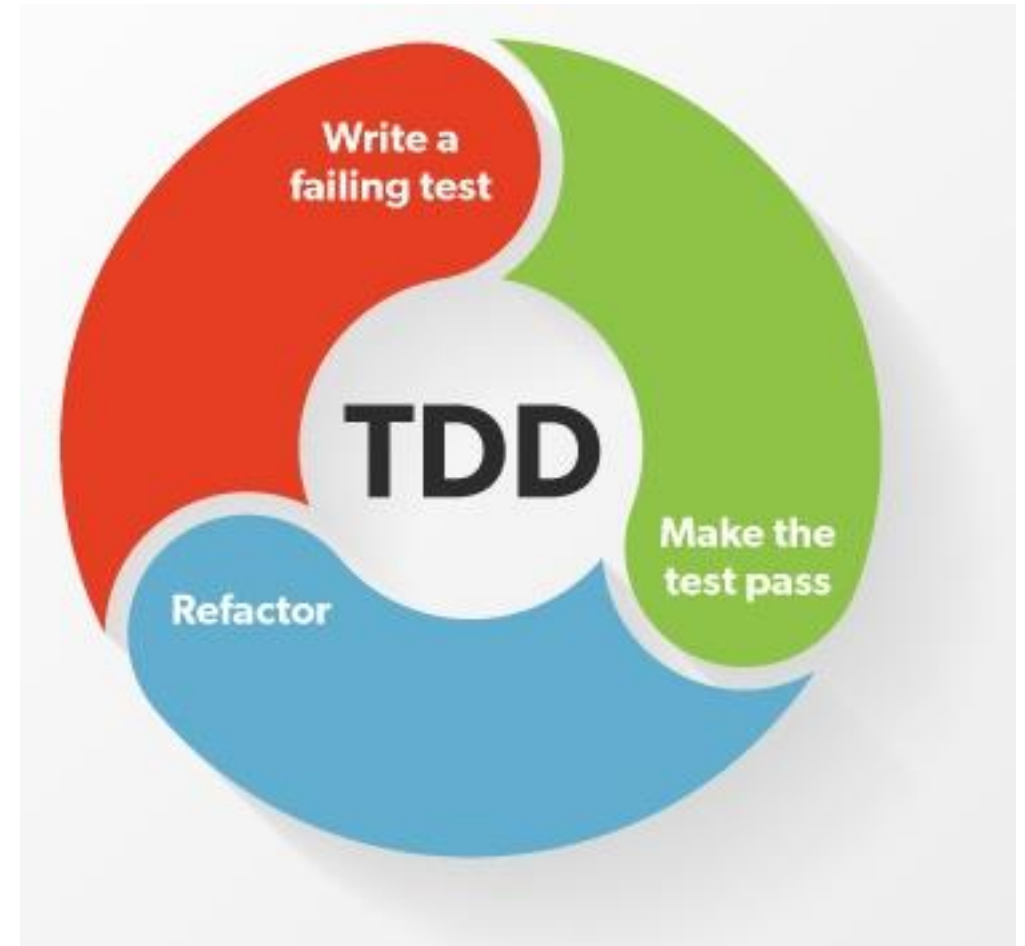
Test Driven Development

- TDD is a process of software development that implements test cases before the software is fully completed
- Aim is to speed up the software development process
- Focus is on simplicity – develop simple clear tests and write only the minimal code to make the test pass
- Once tests pass, code can be refactored for efficiency or readability and tests can be repeated



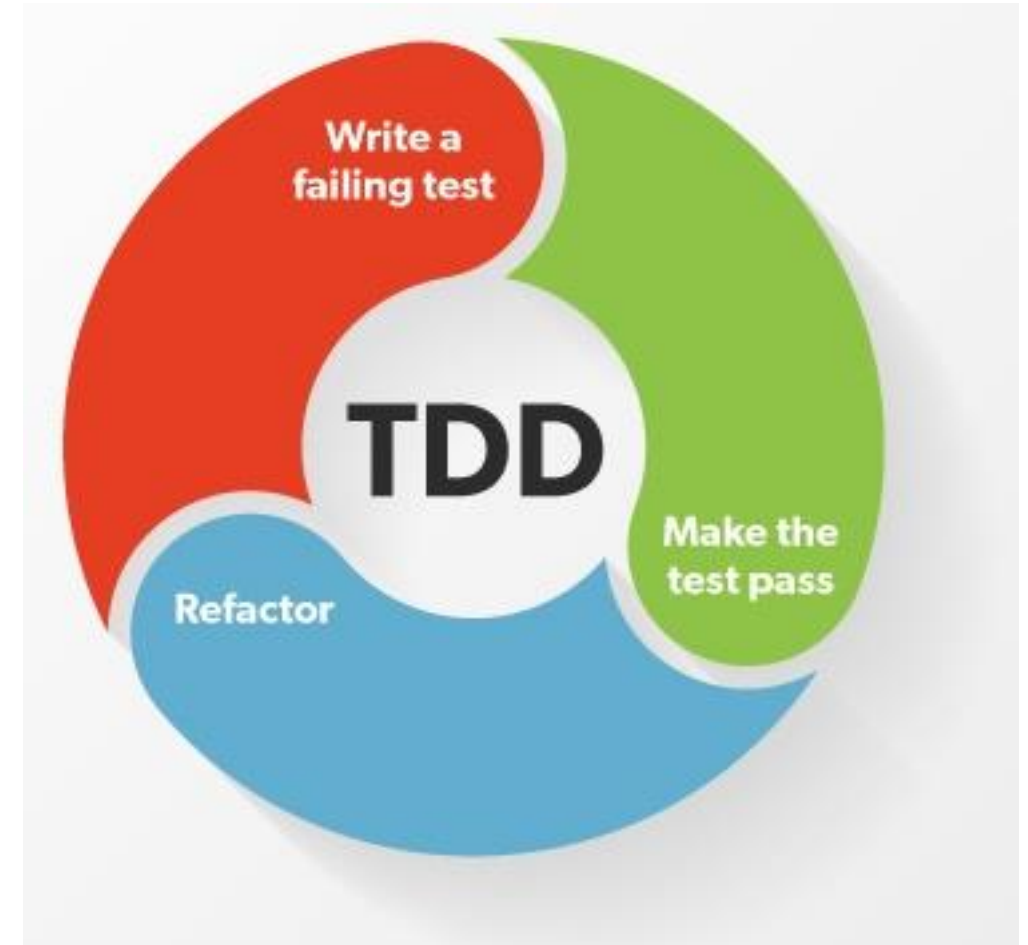
Test Driven Development

1. Create a test
2. Run all tests. The new test should fail for expected reasons
3. Write the simplest code that passes the new test
4. Continue until all tests have passed
5. Refactor code as needed
6. Repeat steps 1-5 as needed



Case Study : ISBNValidator

- An application to check the validity of an ISBN
 - ISBN – International Standard Book Number
 - Introduced 1970 by ISO
 - Originally a 10-digit code
 - Now (since 2007) a 13-digit code



Checking a 10-digit ISBN

ISBN	1	3	9	8	5	2	7	4	9	1
Multiply	10	9	8	7	6	5	4	3	2	1
	10	27	72	56	30	10	28	12	18	1
Total	264									
/ 11	24.0									



Valid ISBN



Checking a 10-digit ISBN

ISBN	1	3	9	8	5	2	7	4	9	2
Multiply	10	9	8	7	6	5	4	3	2	1
	10	27	72	56	30	10	28	12	18	2
Total	265									
/ 11	24.09									



Invalid ISBN



Checking a 13-digit ISBN

ISBN	9	7	8	1	3	9	8	5	2	7	4	9	2
M'ply	1	3	1	3	1	3	1	3	1	3	1	3	1
	9	21	8	3	3	27	8	15	2	21	4	27	2
Total	150												
/ 10	15.0												



Valid ISBN



Checking a 13-digit ISBN

ISBN	9	7	8	1	3	9	8	5	2	7	4	9	3
M'ply	1	3	1	3	1	3	1	3	1	3	1	3	1
	9	21	8	3	3	27	8	15	2	21	4	27	3
Total	151												
/ 10	15.1												



Invalid ISBN

